

Exercise 1.

Check that the relation $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$ does indeed follow from the cut definition by showing
(a) that there is a unique positive real number $\sqrt{n}$ whose square is n , for any integer $n > 0$

(b) that if we define $a \cdot b$ for positive real numbers in the obvious way—you may need to take a little care in doing this— $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$

(c) (Not as difficult as it looks . . .) Let $x_1, x_2, x_3, \dots$ be a sequence of real numbers (defined by cuts) which is bounded above, that is, there exists M such that $x_i < M$ for all i . Prove that there exists M_0 (least upper bound) such that:

(1) $x_i \leq M_0$ for all i ;

(2) if $y < M_0$, then for some i , $x_i > y$.

(d) Define a real number to be an infinite decimal, that is, a series of type $x = a + .a_1a_2a_3 \dots$, where a is an integer and the a_i s are numbers between 0 and 9. In other words, x is the sum of the series $a + a_1/10 + a_2/10^2 + a_3/10^3 + \dots$

What problems arise in devising a rule for adding such numbers?

ANSWER:

A. Let split the rationals into 2 sets L and R . For a positive integer n ,

$$L_n = \{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < n\}$$

$$R_n = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < n\}$$

This is the cut for root n . L has the set of all negative rationals and the right has the set of all positive rationals. Every rational number lies in one of the two sets unless $q^2 = n$ or if it is a non square n then there is no gap.

Let the cut be x .

Since $0 \in L_n$ and there are positive rationals with $q^2 < n$ then the cut is to the right so $x > 0$.

If $x^2 < n$, then there would be a rational y , where y would be more than x while still satisfying that the square of the y would be less than n which is contradictory to the fact that x is supposed to be the cut.

If $x^2 > n$, then there would be another rational z , where the z would be less than x and z^2 would be more than n which is also contradictory to the fact that x is supposed to be the cut.

Hence proving $x^2 = n$.

$$\begin{aligned} \text{If } a \text{ and } b > 0, \text{ and } a^2 = b^2 = n, \\ &= (a + b)(a - b) \\ &= a^2 - b^2 \\ &= 0 \end{aligned}$$

Since $a+b > 0$ then $a - b = 0$ which means $a = b$.

So thus there is one positive real number whose square is n

B. For positive cuts a and b ,

Let ab be the positive real number determined by all rational products xy in the lower set.

Thus, $L_{ab} = \{r \in \mathbb{Q} : r \leq xy \text{ for positive rationals } x < a, y < b\}$

So $x < a, y < b, x, y > 0$

$$\text{Let } a = \sqrt{2}, b = \sqrt{3}$$

$$\text{Then } (ab)^2 = a^2b^2 = (\sqrt{2})^2 \times (\sqrt{3})^2 = 2 \times 3 = 6$$

Since we know by part a that there is a unique positive real number whose square is 6 which is root 6 and that $ab > 0$.

$$\sqrt{2} \times \sqrt{3} = \sqrt{6}$$

C. Let each x_i be represented by a cut (L_i, R_i) where $L = \{q \in \mathbb{Q}(\text{rationals}) : q < x_i\}$ and $R_i = \{q \in \mathbb{Q} : q > x_{i+1}\}$. Since the sequence of x 's is bounded above by M , i.e. $x_i < M$ for all i , let $U = \{q \in \mathbb{Q} : q > x_i \text{ for all } i\}$ which is nonempty since it is at least all the numbers larger than M and obviously has no greatest element. Let $L = \mathbb{Q} \setminus U$, which is non empty since it is at least the numbers less than x_1 and has no greatest element, which can be proved by contradiction.

If L has greatest element g , for some index $k < x_k$, and any number larger than g is in U . Because g is the greatest element of L , any rational number larger than g must belong to U . In particular, any rational $q > g$ satisfies $q > x_i$ for all i . But since $g \leq x_k$, if x_k is rational and $x_k > g$, then x_k in U by the above reasoning. If x_k is irrational, we can pick a

rational q with $g < q < x_k$. Take q to be either x_k (if rational and $> g$) or a rational between g and x_k (if x_k irrational or $x_k = g$). Then $q > g$, so by the property of g being the greatest element of L , we must have q in U . But q in U means $q > x_i$ for all i . In particular, $q > x_k$. This contradicts $q \leq x_k$.

The pair (L, U) forms a cut, let M_0 be the real number corresponding to this cut.

- a. For any i and any rational $q < x_i$, we have q in $L_i \subseteq L$, so $q < M_0$. Thus $x_i \leq M_0$.
 - b. If $y < M_0$, there exists rational q with $y < q < M_0$. Then q in L , so q not in U , meaning there exists some i with $x_i \geq q$. Since $q > y$, we have $x_i > y$.
- D. Attempting to add two numbers represented in this way which have endlessly repeating digits in decimal form but sum to an integer, like $\frac{2}{3} + \frac{1}{3} = .333... + .666... = .999... = 1$. It seems as though the rule would not accept the leftmost equivalence. Additionally, infinite sequences of carried sums would be difficult to handle, like in $.666... + .666...$, where starting from the rightmost digit to carry ones is impossible, and if working from the left it is impossible to know that there will be a carry from the sum of the digits to the right.

Exercise 2.

(a) A shop contains an infinite number of pairs of socks $S_1, S_2, \dots$. I want to choose one sock from each pair.

(1) Why will I need the Axiom of Choice to do it?

(2) Why will I not need the axiom if I am dealing with pairs of shoes?

Answer:

(1) Each pair of socks $S_1, S_2, \dots$ consists of two identical (indistinguishable socks).

To choose one sock from each pair one must make an arbitrary selection from each S_i as there is no rule that distinguishes one sock from another within a pair and since there are infinitely many pairs one must make infinitely many arbitrary choices simultaneously which is what The Axiom of Choice guarantees i.e the existence of a choice function f such that $f(S_i)$ belongs to S_i for all i . Without the axiom there is no way to define or guarantee the existence of such a function

(2) Shoes in a pair are distinguishable as the left and right shoes look different from one another. One can define a choice function explicitly i.e $f(S_i) = \text{left shoe of pair } S_i$ and since this rule specifies a unique element from each pair without any arbitrary choices, no axiom of choice is needed and the function is constructively defined.

(b) Show using the Law of the Excluded Middle that there exist numbers x, y such that x, y are irrational but $x \cdot y$ is rational. [Hint: Start with $x = \sqrt{2}$.] What would be an intuitionist view of this argument?

Answer:

Consider $x = y = \sqrt{2}$ then $x^y = (\sqrt{2})^{(\sqrt{2})}$. Applying the law of the excluded middle, either $(\sqrt{2})^{(\sqrt{2})}$ is rational or it is irrational.

Case 1: $(\sqrt{2})^{(\sqrt{2})}$ it is rational then we are done with $x = y = \sqrt{2}$ (both irrational and x^y is rational)

Case 2: $(\sqrt{2})^{(\sqrt{2})}$ is irrational. Then set $x = (\sqrt{2})^{(\sqrt{2})}$ (which is irrational by assumption) and $y = \sqrt{2}$ (irrational). Then $x^y = ((\sqrt{2})^{(\sqrt{2})})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2} \cdot \sqrt{2})} = 2$ Which is rational

Intuitionist view: An intuitionist would object to this proof because it does not tell us which view it exactly holds. We proved that irrational x, y with x^y rational exists but we cannot point to a specific pair. The proof is non-constructive as it relies on asserting that $(\sqrt{2})^{(\sqrt{2})}$ is either rational or irrational without determining which. An intuitionist would demand that an existence proof provide an explicit example.

(c) (Bolzano-Weierstrass theorem—hard!). Let $x_1, x_2, \dots$ be an infinite sequence of numbers in the unit interval $[0, 1]$.

(1) Show that there is a subsequence $x_{i_1}, x_{i_2}, \dots$ which tends to a limit x .

(2) What, from the intuitionist point of view, has gone wrong here?

Answer:

(1) Divide the interval into two halves ; $[0, 1/2]$ and $[1/2, 1]$. At least one of these half-intervals contains infinitely many terms of the sequence since if both of the intervals contained finitely many then the sequence would be finite which is a contradiction. Pick the half that contains infinitely many terms and let us call it I_1 and pick x_{i_1} belonging to I_1 .

Bisect I_1 into two quarters . Again at least one sub-interval contains infinitely many terms of the remaining sequence and let us call it I_2 .

Continue inductively till stage k we have a closed interval I_k of length $1/2^k$ and we pick x_{i_k} belonging to I_k with $i_1 < i_2 < \dots < i_k$.

The nested intervals $\dots I_3 \subset I_2 \subset I_1$ with lengths close to 0 determine a unique point x . Since x_{i_k} belongs to I_k and the length of I_k is $1/2^k$ tends to 0 , we have x_{i_k} tends to x .

(2)

- The law of the excluded middle is used when we assert at least one of the two halves to contain infinitely many terms. This is non-constructive as we assert that

the two things must be true without specifying which. The same reason applies for each step as we keep bisecting intervals.

- Infinitely many choices are made. At each stage we choose a half interval and a sequence element from. This constitutes an infinite sequence of choices.
- The limit point x is not computable. We know that the nested intervals converge to some x but we cannot in general compute x because we cannot determine which half-interval will be selected at each stage.
- An intuitionist would require a constructive procedure and the classical proof is not constructive.

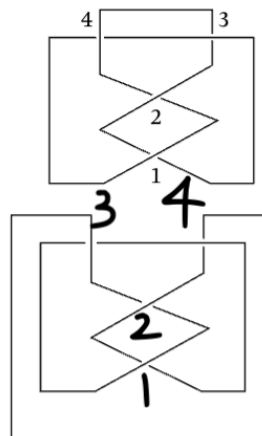


Fig. 9 Two knots, each with four crossings. The crossings on knot 1 are labelled 1, 2, 3, 4 (and obviously those on knot 2 could be labelled similarly).

Exercise 3.

(a) Show that the two knots shown (Fig. 9) are equivalent directly.

Knot 1, going around the closed curve:

- At crossing 1: strand goes over
- At crossing 2: strand goes under
- At crossing 3: strand goes over

- At crossing 4: strand goes under

Knot 2, going around its closed curve in the same oriented direction (labelled it manually):

- At crossing 1: strand goes over
- At crossing 2: strand goes under
- At crossing 3: strand goes over
- At crossing 4: strand goes under

Both knots have the same pattern at 4 crossings, and the order in which the crossings are met as you move around the closed curve is the same in both diagrams. Therefore, the two diagrams are equivalent. Since a knot diagram is fully determined by its crossing data and the order crossings are encountered, the two diagrams represent the same knot with no moves required.

(b) Show that they are equivalent using Reidemeister moves. How many have you used?

Step 1: Type III move at crossings 1, 2, 3, slide the strand passing through crossing 1 past the intersection point of crossings 2 and 3. Crossing count remains 4. The upper portion of the diagram shifts toward knot 2's layout.

Step 2: Type II move at crossings 3, 4, pull the strand over, temporarily removing crossings 3 and 4 so reducing to 2 crossings. Then introduce them in the new position matching knot 2's lower structure, which repositions the bottom strands.

Step 3: Type III moves at the remaining crossings, slides the final strand through to complete the transformation, then matches the diagram into exactly knot 2's layout.

So, there are 3 Reidemeister moves with one Type II and two Type III.